

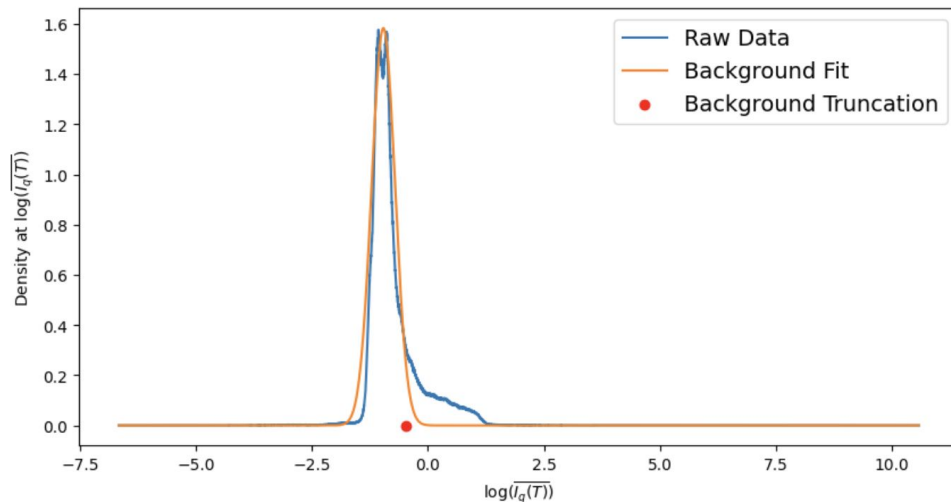
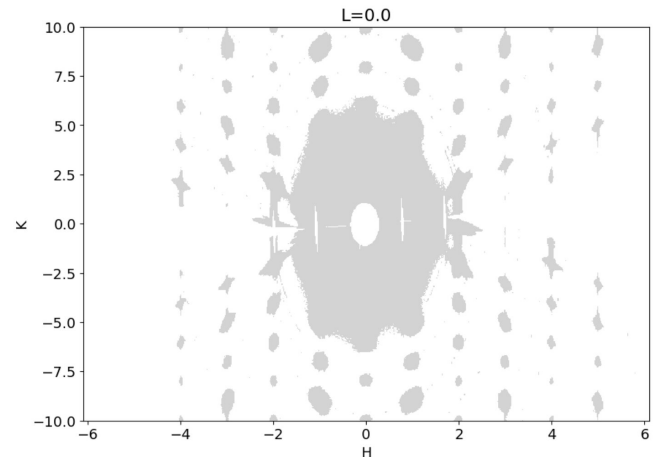
Advanced Clustering

Goals For This Week

- Plot Log Intensity Distribution with our data
- Find the correct amount of clusters
- Automated selections of parameters
 - Amount of clusters and rescaling method (mean vs. z-score)

Log Intensity Distribution

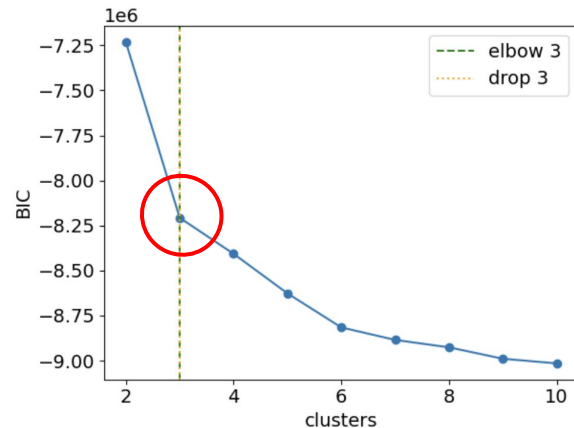
- Pixel intensity is averaged over 24 temperatures, then plotted
- The cut off is the red dot/background truncation
- Pixels brighter than the cut off are kept (the rest is noise)



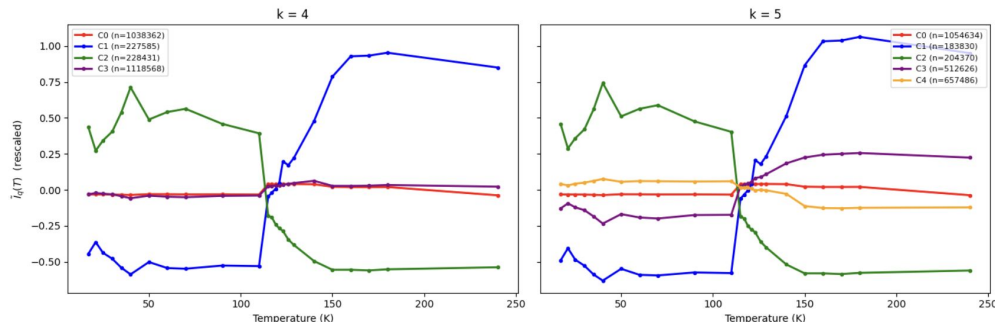
Finding The Correct Amount of Clusters

- Used the BIC score with the elbow method
 - BIC measures how well a statistical model fits the data
- Elbow and greatest drop agree that there should be 3 clusters
- Show the trajectory plots to visualize

greatest-drop k=3, elbow k=3

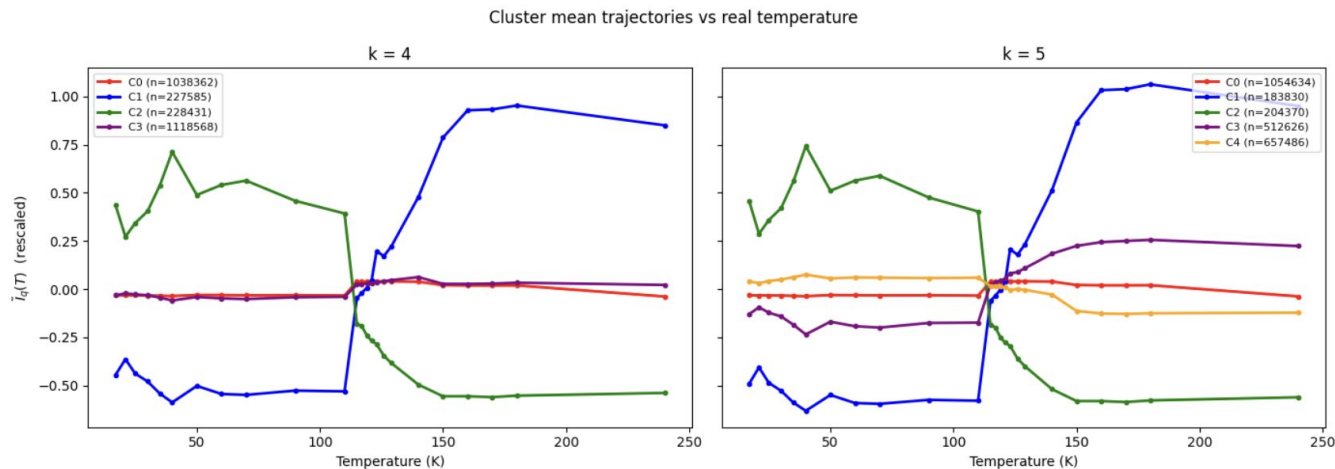


Cluster mean trajectories vs real temperature



How do you pick?

- The drop between 3 and 4 is steeper than 4 and 5, so BIC leans toward 4 clusters
- Either way, both order parameters show up
- Comes down to judgement



Automated Parameter Tuning

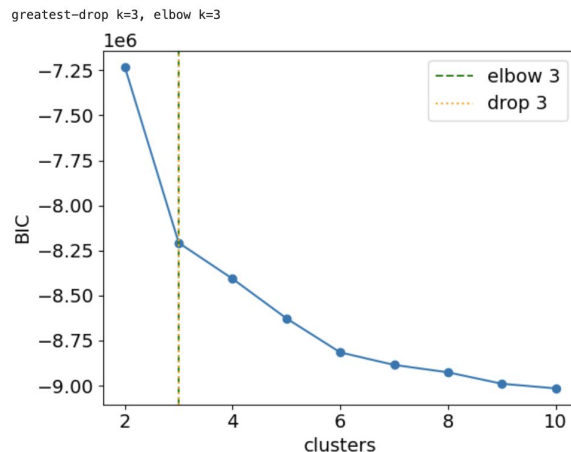
- Rescales the data using both methods (mean vs z-score) then calculates silhouette score
 - Measures how well-separated the clusters are
 - -1 to +1, higher = better
- BIC score picks the number of clusters
 - Greatest drop
 - Elbow method
- Automatically picks the rescale method and k

Automated Parameter Tuning Results

- Selected mean rescaling method since it had the highest silhouette score
- Found that the elbow was at k=5 and the greatest drop was at k=4

```
[mean] silhouette score = 0.0840 (higher = better separated)
[zscore] silhouette score = 0.0631 (higher = better separated)
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```
Selected rescale method (by silhouette): mean
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Next Steps

- Automate the covariance type and threshold cutoff parameters
- CPU/GPU comparison, time analysis
- Build an AI agent to automate clustering decisions